



What the finite-horizon bias bounds tell you

Categorical MGW shared exclusions · five exact formal families

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8 September 2026

For a **fixed finite categorical MGW atom**, a **fixed raw horizon** h and **iid complete source-and-target rows**, five locally accepted Lean statements bound mean truncation error. Within-row dependence is allowed. Each block uses a fresh anchor and h further rows from the same law. Information is in nats.

Let $p(z)$ be the complete-row population PMF, θ the atom and B_h the [Section 3 prefix statistic](#). Define $e_h = \theta - \mathbb{E}[B_h]$, opposite in sign to conventional bias. At each anchor with $p(z) > 0$, let $a_z = P(\text{source query})$, $v_z = P(\text{target match})$ and $q_z = P(\text{query and target match})$. The query ORs source collections and ANDs matches within each collection.

The supported population moments $K_a = \sum_{z:p(z)>0} p(z)/a_z$, $K_q = \sum_{z:p(z)>0} p(z)/q_z$ and $K_T = \sum_{z:p(z)>0} p(z)/v_z$ give

$$-\frac{K_q - K_T}{h+1} \leq e_h \leq \frac{K_a - 1}{h+1}, \quad |e_h| \leq \frac{K_q - 1}{h+1}.$$

For **any member collection** C of the chosen node, let K_C and K_{CT} be the positive population support counts of its source projection and source-target projection. Then

$$-\frac{K_{CT} - K_T}{h+1} \leq e_h \leq \frac{K_C - 1}{h+1}, \quad |e_h| \leq \frac{K_{CT} - 1}{h+1}.$$

A justified population floor $q_z \geq \eta$ for **every** anchor with $p(z) > 0$, with $0 < \eta \leq 1$, gives the potentially faster bound

$$|e_h| \leq \frac{(1 - \eta)^{h+1}}{(h+1)\eta}.$$

These are **sufficient fixed-horizon planning bounds** for known population inputs. Small source-target projections may improve the support bound. Observed category counts or minimum observed frequencies cannot replace population support counts or floors without further evidence.

The bounds can be conservative: constant sources have zero signed error even with large K_q . Overlapping queries need not be projection fibers: $K_q = 4/3$ is possible. For one fair source independent of a fair target, $\mathbb{E}[B_1] = 1/4$ while $\theta = 0$. A positive short-prefix mean alone does not show population information.

The bounds control **mean truncation error**. Sampling variation, confidence coverage, stopping, implementation accuracy and utility remain open. Binning changes the estimand to the resulting categorical law. Same-sample fitting, serial dependence, continuous PID and hyperbolic PID need separate arguments. For prediction or a one-source relation, consider direct task loss or MI.

See the derivation and four worked examples, [theorem map](#), and defining MGW v5 [paper](#). Local formal acceptance is separate from public integration and hosted replay. No novelty or superiority is claimed.

